

Figure 13: Swap strategy in a graph with two connected lines each with five vertices. The blue SWAP gates show the first swap layer. The qubit order in each line fully reverses after five steps and full connectivity is reached after four steps.

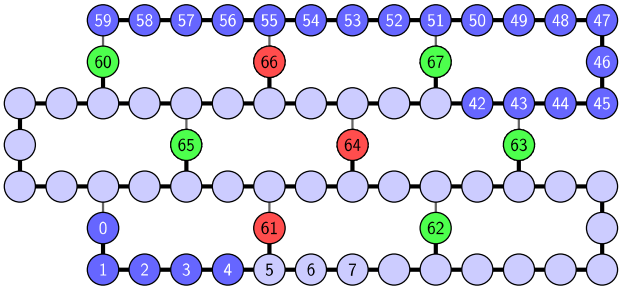


Figure 14: A three by three heavy-hex coupling map. The longest line ranges from qubits 0 to 60. By contrast, the blue qubits are part of the longest-line of the *unfolded* heavy-hex graph with $l = 60$. The dark blue nodes indicate the tails with five and $t = 18$ qubits. The red and green nodes indicate the A and B qubits, respectively. The grey edges are the edges that are removed to unfold the heavy-hex graph.

A.3 Heavy-hex

A heavy-hex coupling map has a mixture of degree-two and degree-three nodes. Its qubits are placed on the edges and vertices of hexagons [25]. Each hexagon therefore has 12 qubits. Here, we focus on $i \times j$ heavy-hex graphs which have i rows and j columns of hexagons, as exemplified by the 3×3 heavy-hex graph in Fig. 14. The total number of qubits is $n = 5ij+4(i+j)-1$ and the length of the longest line in the graph is $l_{\max} = 4(ij + i + j) + 1$. The length of this line is related to the total number of qubits by $l_{\max} = 4[n(i,j)+i+j+1]/5+1$ which, to leading order, scales as $4n/5$. Furthermore, $l_{\max}$ is bound from above by

$$\frac{4}{5}n + \frac{4}{5}\sqrt{n} + 1 \tag{8}$$

when the grid of hexagons is approximately square, i.e. $i \sim j$. These preliminaries allow us to formulate the following Lemma for the heavy-hex swap strategy.

Lemma 3. *For the approximately square heavy-hex grid of size n, there exists a swap strategy that reaches full connectivity in less than $n+\sqrt{n}+61$ swap layers.*

Proof. We prove Lemma 3 by unfolding the heavy-hex graph along a line of length l where $l \mod 4 = 0$, see Fig. 14. To unfold, we delete one edge connected to the nodes not in the longest line. The result is a line graph with an optional additional node connected to every fourth node, see Fig. 15. The graph additionally has a tail of 5 vertices on one side and a tail of t vertices on the other side, such that $t = 2 \mod 4$, see Fig. 14. Here, t depends on the width of the heavy-hex graph. We will prove that Lemma 3 holds for any graph of this kind.

The proof applies a line swap strategy on the unfolded heavy-hex graph modified such that at any time some qubits remain in the positions of the additional nodes without moving along the line. We divide the process into five iterations. In each iteration a qubit either moves along 1/4 of the line or waits in one of the adjacent nodes. If every qubit only enters a waiting state once during the complete process, all qubits will have completed l steps of the simple line strategy after five iterations, leading to full connectivity. The difficulty then lies in ensuring that every qubit is swapped into a waiting position at most once. We divide the additional nodes into two sets A and B spaced apart by eight nodes in the line, see Fig. 15. We number the edges in the line by their position and define four swap layers to reach full connectivity.

- S_1: All odd-numbered edges in the line.
- S_2: All even-numbered edges in the line.
- S_3: All edges connected to vertices in group A.
- S_4: All edges connected to vertices in group B.

We claim that the swap strategy in which we

1. alternate between S_1 and S_2 $k-7$ times starting with S_1, and
2. apply S_4 once, and
3. alternate between S_1 and S_2 7 times starting with S_2, and
4. apply S_3 once, and
5. repeat Steps 1-4 five times,

reaches full connectivity on the unfolded heavy-hex if

$$k = \frac{l}{4} - \left(\frac{l}{4} \mod 8\right) + 10. \tag{9}$$

Here, k is chosen such that $k \mod 8 = 2$ and $k > l/4$ so that every qubit travels the full line with four iterations of steps 1. and 3. Steps 2. and 4. swap qubits in and out of nodes A and B, respectively. Steps 1. to 4. require $k+2$ swap layers so that after five iterations the total number of swap layers is $5k + 10$ which is thus bounded from above by $n + \sqrt{n} + 61$ as seen by injecting Eq. (8) in Eq. (9) and conservatively assuming $l/4 \mod 8 = 0$.

As argued above, it suffices to show that any particular qubit will remain in a position corresponding to groups A or B for at most one iteration. We assign

each node along the line to one of eight evenly-spaced groups V_i with $i \in \{0, \ldots, 7\}$. The set of all vertices is thus partitioned into ten sets, A, B and V_i, see Fig. 15. We now examine how the qubits switch groups during steps 1. to 4. Note that during each iteration k steps of the line strategy are executed and k was chosen, such that $k \bmod 8 = 2$.

We first ignore groups A and B. Since k is even and we begin by applying SWAP gates to odd numbered edges in the line, i.e. S_1, qubits moving towards the left will switch from group V_i to V_{i-2} and qubits moving towards the right will move from V_i to V_{i+2}. If a qubit reaches the end of the line during an iteration it will either

- switch direction from left to right and switch groups from V_i to V_{7-i}, or
- switch direction from right to left and switch groups from V_i to V_{7-i+4}.

This behavior results from $k \bmod 8 = 2$. The movement pattern thus undergone by individual qubits is depicted in Fig. 16.

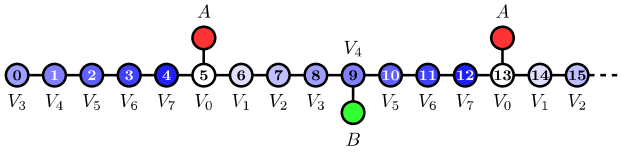


Figure 15: The unrolled heavy-hex graph with numbered nodes. The nodes are divided into ten subsets A (in red) B (in green) and V_i for $i \in \{0, \ldots, 7\}$ (from white to blue).

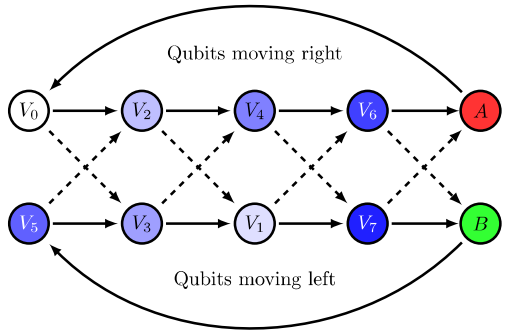


Figure 16: Qubit movement pattern during the heavy-hex swap strategy. Each node shows one of the ten categories of nodes defined in Fig. 15. The arrows show how the qubits change group after an iteration of steps 1. through 4. Solid and dashed lines show qubits that were and were not reflected at line ends during the iteration, respectively. Note that a dashed arrow from V_6 to V_5 is not shown.

We now consider groups A and B. During one iteration qubits positioned in groups A and B will switch to groups V_0 and V_5, respectively. From the outlined strategy, qubits only arrive in group A if they ended the previous iteration in a group V_i with

$$\begin{aligned} i + 2 = 0 \bmod 8 \quad &\implies i = 6 \\ \text{or} \quad 7 - i = 0 \bmod 8 \quad &\implies i = 7. \end{aligned}$$

Since the swap layer swapping with group B is executed after $k - 7$ layers of S_1 in alternation with S_2 and nodes of group B are connected with group V_4, a qubit can only arrive in group B if it was initially positioned in a group V_i with

$$\begin{aligned} i - 2 + 7 = 4 \bmod 8 \quad &\implies i = 7 \\ \text{or} \quad 7 - i + 4 + 7 = 4 \bmod 8 \quad &\implies i = 6. \end{aligned}$$

In either case, a qubit can only arrive in groups A or B if it was previously positioned in group V_6 or V_7. All possible movements of qubits among groups from one iteration to the next are thus captured by Fig. 16 and it is clear that within five iterations each qubit will only visit A or B at most once. This concludes the proof. □

B Circuit depth and CNOT gate count

We now investigate how the number of CNOT layers and gates in QAOA circuits, transpiled using the pass described in Sec. 2.1, scale with problem size. Here, $G_0 = (V, E_0)$ is either a heavy-hex or an η-dimensional grid coupling map and $\{S_i\}$ with $i \in \mathbb{N}$ is a swap strategy compatible with G_0. The transpiled circuit has alternating layers of R_{ZZ} and SWAP gates applied on edge sets E_i and S_i, respectively, see Fig. 17. We determine the number of CNOT gates and layers by counting gates and layers in E_i and S_i taking into account gate cancellations across layers.

In every layer E_i we apply R_{ZZ} gates on the edges in E_i. The number of edges in E_i thus determines the number of R_{ZZ} gates. Here, E_i is the set of edges in the hardware coupling map (i.e. $E_i \subseteq E_0$) which give new qubit connections after applying the swap layer S_{i-1}. Edges which may give new qubit connections after S_{i-i} are in the neighbourhood of swapped edges, i.e.

$$E_i \subseteq \bigcup_{e \in S_{i-1}} N(e), \tag{10}$$

as exemplified in Fig. 18. Since the SWAP gates in a swap layer S_i are executed in parallel S_i never contains two neighboring edges, i.e. $e \notin N(e')\ \forall\ e, e' \in S_i$. Equation (10) therefore implies $E_i \subseteq E_0 \setminus S_{i-1}$.

Carefully positioned CNOT gates across E_i and S_i may cancel. We therefore position all R_{ZZ} gates of E_i that are applied across edges contained in S_i at the end of E_i. This allows us to simplify two CNOT gates and implement SWAP$\cdot$R$_{\text{ZZ}}(\theta)$ with three CNOTs and a $R_Z(\theta)$ gate, see Fig. 19. The number of CNOT layers $L_{cx}(E_i, S_i)$ required to implement E_i and S_i is